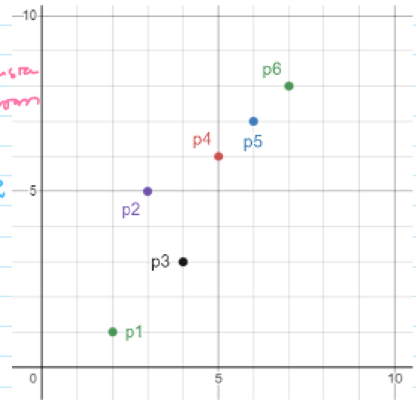
# PCA Example

"Data Set":

| F1 | F2 | |
|---|---|---|
| 2 | 1 | P1 |
| 3 | 5 | P2 |
| 4 | 3 | P3 |
| 5 | 6 | P4 |
| 6 | 7 | P5 |
| 7 | 8 | P6 |

dimension of room

F2




F1

→ # of room

Step 1: Compute Covariance Matrix

Mean (F1) = 2+3+4+5+6+7 / 6 = 4.5

Mean (F2) = 1+5+3+6+7+8 / 6 = 5

• $Var(x) = \sum_{i=1}^{n} (x-\bar{x})^2 / n-1$ → Sample

• $cov(x,y) = \sum_{i=1}^{n} (x-\bar{x})(y-\bar{y}) / n-1$

M = Covariance Matrix

$$M_{|F|\times|F|} = \begin{bmatrix} Cov(F1, F1) & Cov(F1, F2) \\ Cov(F2, F1) & Cov(F2, F2) \end{bmatrix} = \begin{bmatrix} Var(F1) & cov(F1, F2) \\ Cov(F1, F2) & Var(F2) \end{bmatrix}$$

$$Var(F1) = (2-4.5)^2 + (3-4.5)^2 + (4-4.5)^2 + (5-4.5)^2 + (6-4.5)^2 + (7-4.5)^2 / 5 = \boxed{3.5}$$

$$Var(F2) = (1-5)^2 + (5-5)^2 + (3-5)^2 + (6-5)^2 + (7-5)^2 + (8-5)^2 / 5 = \boxed{6.8}$$

$$Cov(F1, F2) = (2-4.5)(1-5) + (3-4.5)(5-5) + (4-4.5)(3-5) + (5-4.5)(6-5) + (6-4.5)(7-5) + (7-4.5)(8-5) / 5 = \boxed{4.4}$$

$$M = \begin{bmatrix} 3.5 & 4.4 \\ 4.4 & 6.8 \end{bmatrix}$$

Step 2: Eigen Decomposition $\quad \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$

1. Eigen values $\quad |M_{2\times 2} - \lambda I_{2\times 2}| = 0$

2. Eigen Vector

$$\begin{vmatrix} 3.5-\lambda & 4.4 \\ 4.4 & 6.8-\lambda \end{vmatrix} = 0$$

$$(3.5-\lambda)(6.8-\lambda)-(4.4)(4.4)=0$$

$$23.8-3.5\lambda-6.8\lambda+\lambda^2-19.36=0$$

$$\lambda^2-10.3\lambda+4.4=0$$

$a = 1$
$b = -10.3$
$c = 4.4$

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

$$x=\frac{-(-10.3)+\sqrt{(-10.3)^2-4\times1\times4.4}}{2\times1}$$

$$x=\frac{10.3+\sqrt{88.5}}{2}$$

$\lambda_1$
$= \boxed{9.85}$ "Pc1"

$$x=\frac{-(-10.3)-\sqrt{(-10.3)^2-4\times1\times4.4}}{2\times1}$$

$$x=\frac{10.3-\sqrt{88.5}}{2}$$

$\lambda_2$
$= \boxed{0.45}$ Pc2

Eigen Vector

$$Mv=\lambda v$$

$$Mv-\lambda v=0$$

$$(M-\lambda)v=0$$

For Pc1 = 9.85

$$\begin{pmatrix}3.5-9.85 & 4.4\\ 4.4 & 6.8-9.85\end{pmatrix}v=0$$

$$\begin{pmatrix}-6.35 & 4.4\\ 4.4 & -3.05\end{pmatrix}v=0$$

$$\left(\begin{array}{cc|c}-6.35 & 4.4 & 0\\ 4.4 & -3.05 & 0\end{array}\right)=0$$

$R_1/-6.35 \rightarrow R1$ 1

$$\left(\begin{array}{cc|c}1 & -0.69 & 0\\ 4.4 & -3.05 & 0\end{array}\right)$$

$R_2 - (4.4 R_1) \to R_2$ ②

$$\left(\begin{array}{cc|c} 1 & -0.69 & 0 \\ 0 & 0 & 0 \end{array}\right)$$

$x_1 - 0.69x_2 = 0$

$x_1 = 0.69x_2$

For $x_2 = 1$

$\begin{pmatrix} 0.69 \\ 1 \end{pmatrix}$

Eigen Vector

Eigen Value

$\boxed{Pc1 = 9.85}$

For $Pc2 = 0.45$

$$\begin{pmatrix} 3.5 - 0.45 & 4.4 \\ 4.4 & 6.8 - 0.45 \end{pmatrix} \cdot v = 0$$

$$\left(\begin{array}{cc|c} 3.05 & 4.4 & 0 \\ 4.4 & 6.35 & 0 \end{array}\right) = 0$$

$R_1 / 3.05 \to R_1$ ①

$$\left(\begin{array}{cc|c} 1 & 1.44 & 0 \\ 4.4 & 6.35 & 0 \end{array}\right) = 0$$

$R_2 - 4.4 \times R_1 \to R_2$

$$\left(\begin{array}{cc|c} 1 & 1.44 & 0 \\ 0 & 0 & 0 \end{array}\right) = 0$$

$x_1 + 1.44x^2 = 0$

$x_1 = -1.44x^2$

$x_2 = 1$

$\boxed{\begin{pmatrix} -1.44 \\ 1 \end{pmatrix}}$

Eigen Vector

Pc2

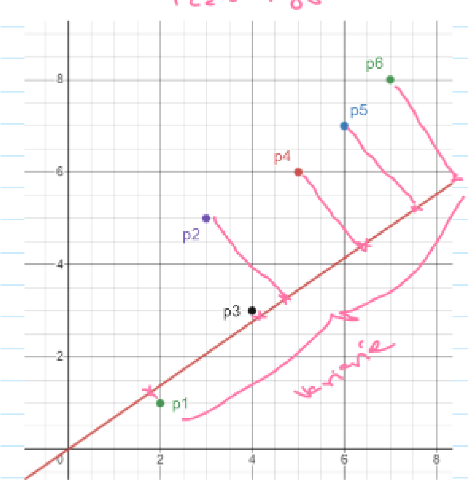
$\boxed{0.45}$

Eigen Value

Eigen Value $\{9.85, 0.45\}$

Eigen Vector $\left\{ \begin{bmatrix} 0.69 \\ 1 \end{bmatrix} \begin{bmatrix} -1.44 \\ 1 \end{bmatrix} \right\}$

$Pc1 = 9.85$



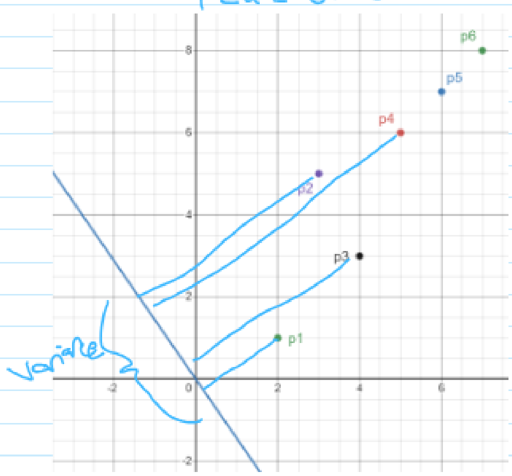

vector

$\boxed{0.69x = y}$

Minimum Loss

• Maximum Variance

$Pc2 = 0.45$



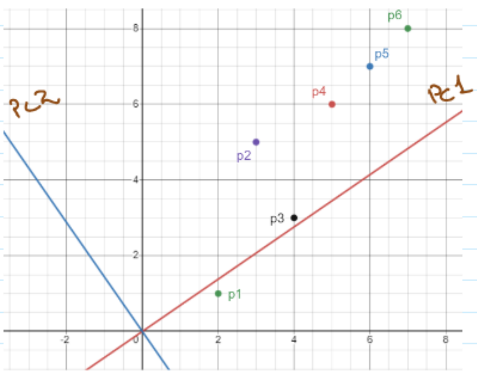

vector

$\boxed{-1.44x = y}$

Small
• Variance

Data loss

Normalized Eigen Vectors $\Rightarrow \left[ \frac{0.69}{\sqrt{(0.69)^2+1^2}}, \frac{1}{\sqrt{(0.69)^2+1}} \right] \left[ \frac{-1.44}{\sqrt{(-1.44)^2+(1)^2}}, \frac{1}{\sqrt{(-1.44)^2+(1)^2}} \right]$




$$\begin{array}{cc} PC1 & PC2 \\ \begin{bmatrix} 0.57 & -0.76 \\ 0.83 & 0.53 \end{bmatrix} \end{array}$$

PC . Matrix

" Data Transformation "

(Data - mean)

$$\underset{6\times 2}{\begin{bmatrix} -2.5 & -4 \\ -1.5 & 0 \\ -0.5 & -2 \\ 0.5 & 1 \\ 1.5 & 2 \\ 2.5 & 3 \end{bmatrix}}$$

(Data-mean) * Normalized Eigen Vectors

$$* \begin{array}{cc} PC1 & PC2 \\ \begin{bmatrix} 0.57 & -0.76 \\ 0.83 & 0.53 \end{bmatrix} \end{array}$$

$F_1 - m_1$ ? $F_2 - m_2$

. captures maximum variance in data

this only selecting for data recovery.

$$\begin{bmatrix} \boxed{\begin{matrix} -4.745 \\ -0.855 \\ -1.945 \\ 1.115 \\ 2.515 \\ 3.915 \end{matrix}} & \begin{matrix} -0.22 \\ 1.14 \\ -0.68 \\ 0.15 \\ -0.08 \\ -0.31 \end{matrix} \end{bmatrix}$$

$F_1'$ w.r.t PC1 $\qquad F_2'$ w.r.t PC2

For data Recovery.

$F_1' * PC1^T$

$$\begin{bmatrix} -4.745 \\ -0.855 \\ -1.945 \\ 1.115 \\ 2.515 \\ 3.915 \end{bmatrix} * \begin{bmatrix} 0.571 & 0.831 \end{bmatrix} \Rightarrow \underbrace{6\times1 \quad 1*2}_{6\times2}$$

$6*1 \qquad 1*2$

$$\begin{bmatrix} -2.7 & -3.9 \\ -0.48 & -0.70 \\ -1.10 & -1.61 \\ 0.63 & 0.925 \\ 1.43 & 2.08 \\ 2.23 & 3.25 \end{bmatrix}$$

$\bar{F_1} \qquad \bar{F_2}$

As we have subtracted mean from each column now we have to add it back

# Recovery Data

$\boxed{\begin{array}{l} m_1 = 4.5 \\ m_2 = 5 \end{array}}$

$F_1 + m_1$ , $F_2 + m_2$

$$\begin{pmatrix} 1.8 & 1.07 \\ 4.01 & 4.3 \\ 3.4 & 3.39 \\ 5.1 & 5.92 \\ 5.93 & 7.08 \\ 6.73 & 8.2 \end{pmatrix}$$

Orignal

$F$

$$\begin{pmatrix} 2 & 1 \\ 3 & 5 \\ 4 & 3 \\ 5 & 6 \\ 6 & 7 \\ 7 & 8 \end{pmatrix}$$